In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
from PIL import Image
```

In [3]:
```
ele=Image.open('C:/Users/ADMIN/elephant.jpeg')
```

In [4]:
```
ele
```

Out[4]:

In [5]:
```
type(ele)
```

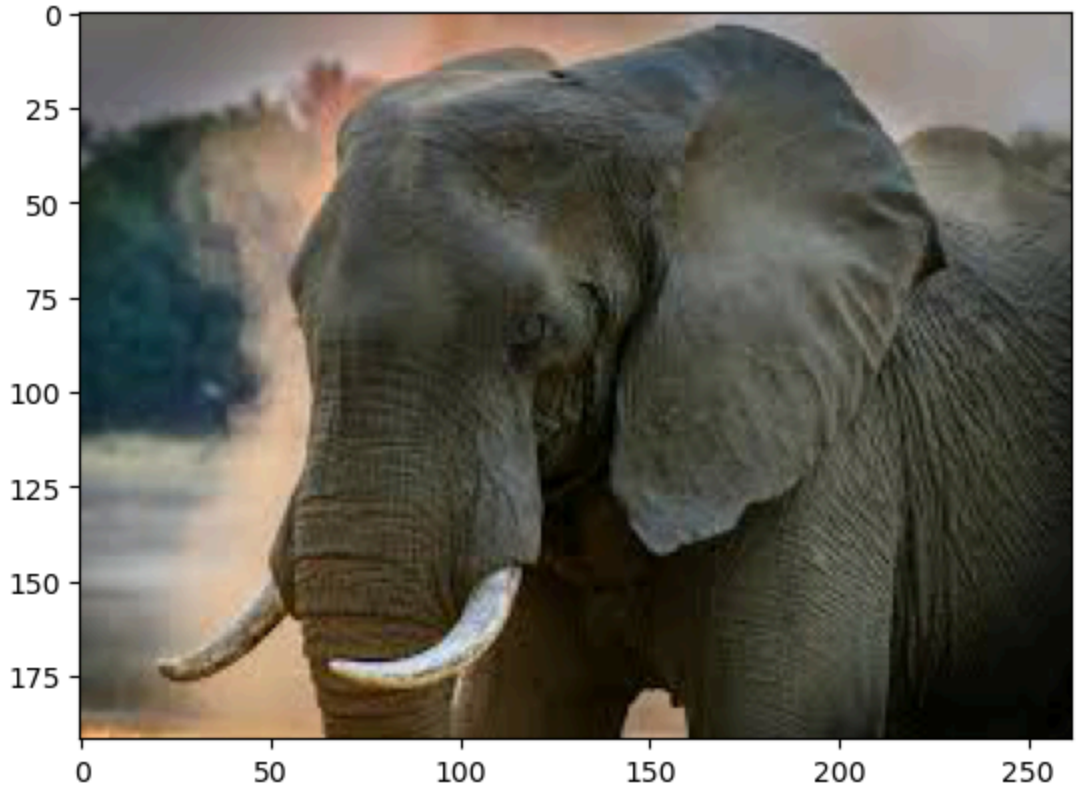
Out[5]:
```
PIL.JpegImagePlugin.JpegImageFile
```

In [6]:
```
ele_array=np.asarray(ele)
ele_array
```

Out[6]:
```
array([[[108, 107, 103],
        [108, 107, 103],
        [108, 107, 103],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       [[108, 107, 103],
        [108, 107, 103],
        [108, 107, 103],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       [[107, 106, 102],
        [107, 106, 102],
        [107, 106, 102],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       ...,

       [[155, 103,  55],
        [169, 119,  70],
        [184, 134,  85],
        ...,
        [  8,   8,   6],
        [ 10,  10,   8],
        [ 11,  11,   9]],

       [[190, 129,  72],
        [203, 142,  85],
        [217, 156,  99],
        ...,
        [ 10,  10,   8],
        [ 10,  10,   8],
        [ 10,  10,   8]],

       [[227, 161, 100],
        [236, 170, 109],
        [247, 181, 120],
        ...,
        [ 10,  10,   8],
        [ 10,  10,   8],
        [  9,   9,   7]]], shape=(192, 262, 3), dtype=uint8)
```

In [7]:
```
plt.imshow(ele_array)
```

Out[7]:
```
<matplotlib.image.AxesImage at 0x2062cdd1160>
```



In [8]:
```
ele_array.shape
```

Out[8]:
```
(192, 262, 3)
```

In [11]:
```
ele_red=ele_array.copy()
ele_red
```

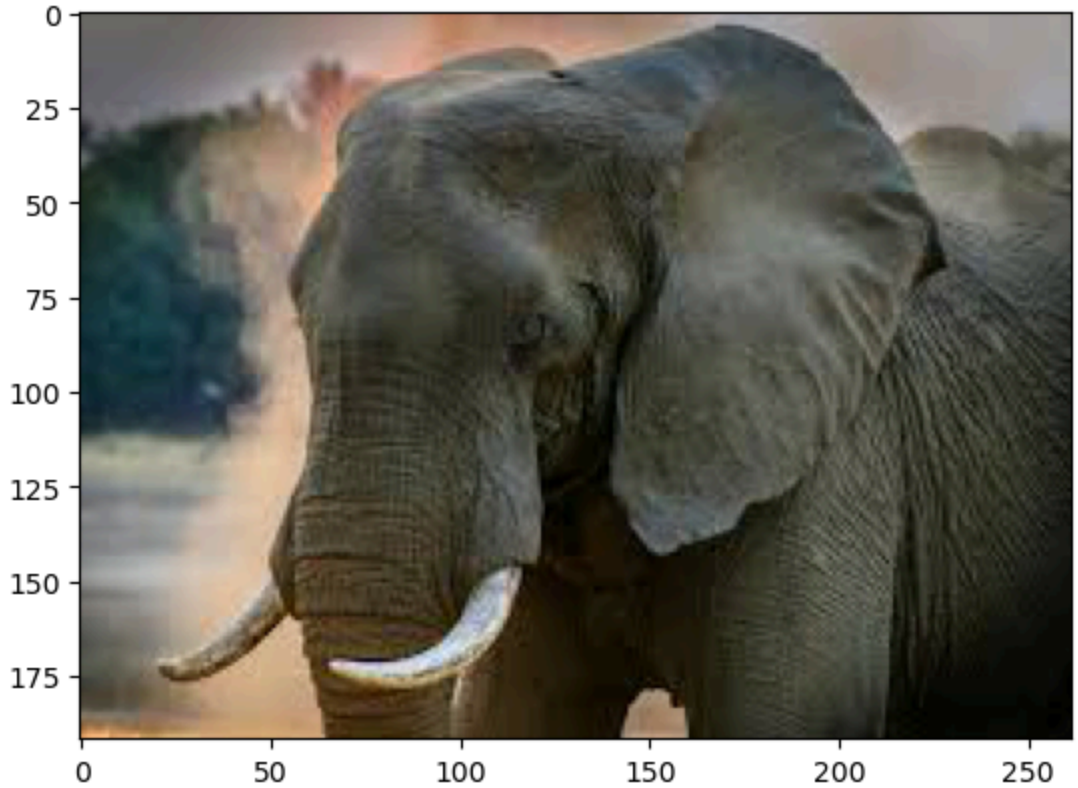
Out[11]:
```
array([[[108, 107, 103],
        [108, 107, 103],
        [108, 107, 103],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       [[108, 107, 103],
        [108, 107, 103],
        [108, 107, 103],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       [[107, 106, 102],
        [107, 106, 102],
        [107, 106, 102],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       ...,

       [[155, 103,  55],
        [169, 119,  70],
        [184, 134,  85],
        ...,
        [  8,   8,   6],
        [ 10,  10,   8],
        [ 11,  11,   9]],

       [[190, 129,  72],
        [203, 142,  85],
        [217, 156,  99],
        ...,
        [ 10,  10,   8],
        [ 10,  10,   8],
        [ 10,  10,   8]],

       [[227, 161, 100],
        [236, 170, 109],
        [247, 181, 120],
        ...,
        [ 10,  10,   8],
        [ 10,  10,   8],
        [  9,   9,   7]]], shape=(192, 262, 3), dtype=uint8)
```

In [12]:
```
ele_array == ele_red
```

Out[12]:
```
array([[[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       ...,

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]]], shape=(192, 262, 3))
```

In [13]:
```
plt.imshow(ele_red)
```

Out[13]:
```
<matplotlib.image.AxesImage at 0x2062e580e10>
```



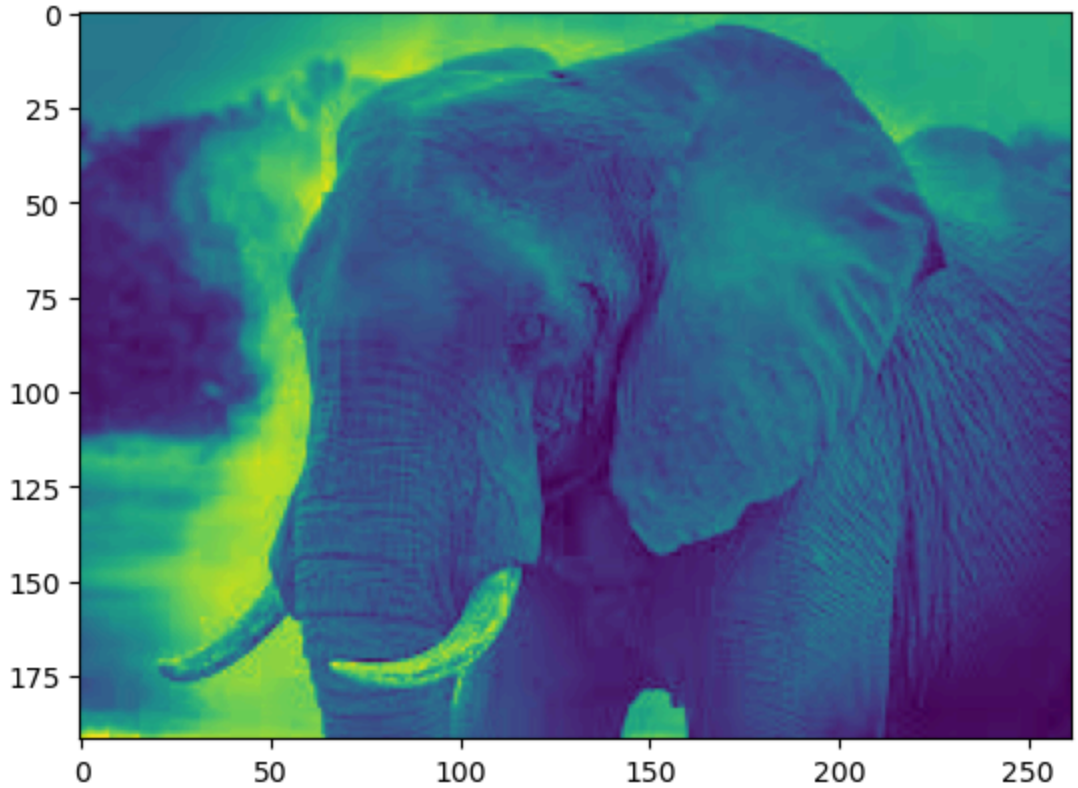
In [14]:
```
ele_red.shape
```

Out[14]: (192, 262, 3)

========== RGB ==========

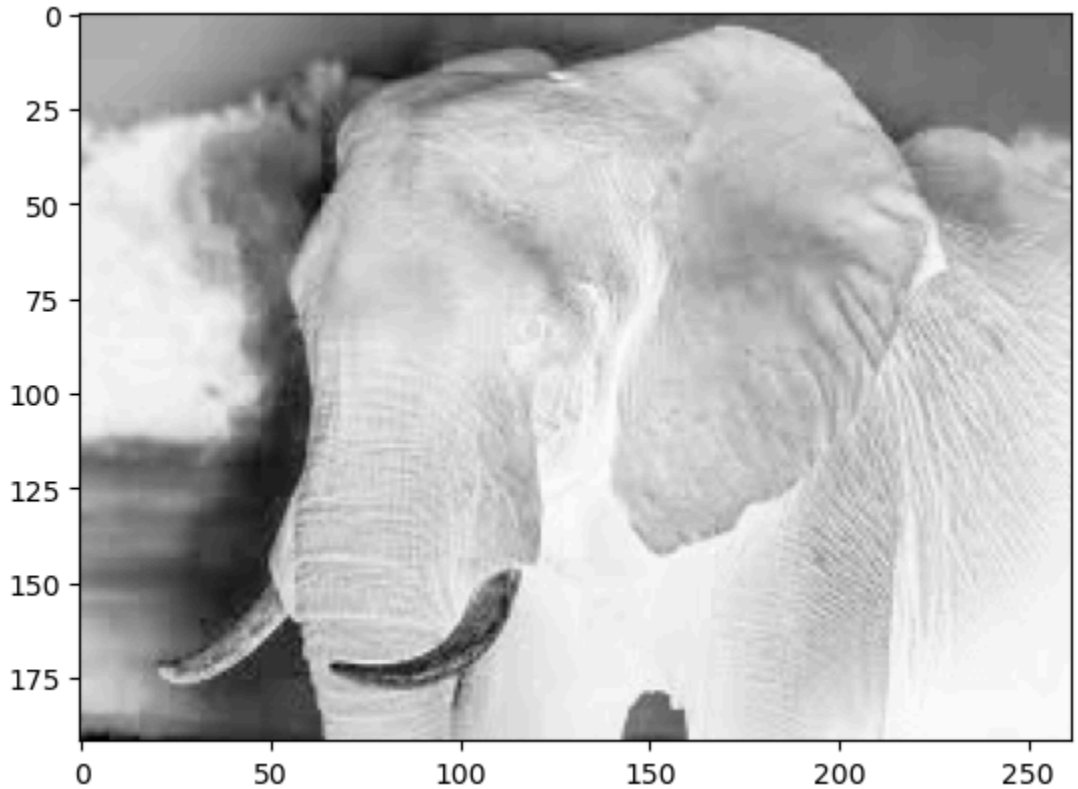
In [15]:
```
plt.imshow(ele_red[:,:,0])
```

Out[15]: <matplotlib.image.AxesImage at 0x2062f3cc910>



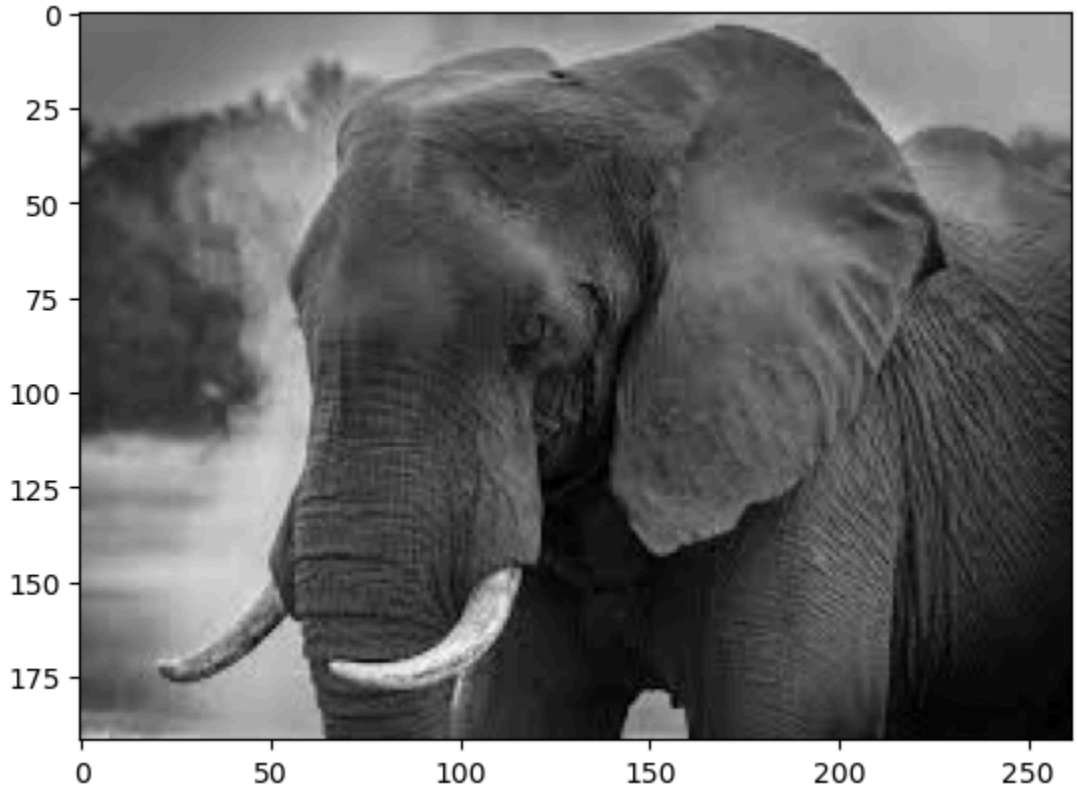
In [16]:
```
plt.imshow(ele_red[:,:,0],cmap='Greys')
```

Out[16]: <matplotlib.image.AxesImage at 0x2062e913890>



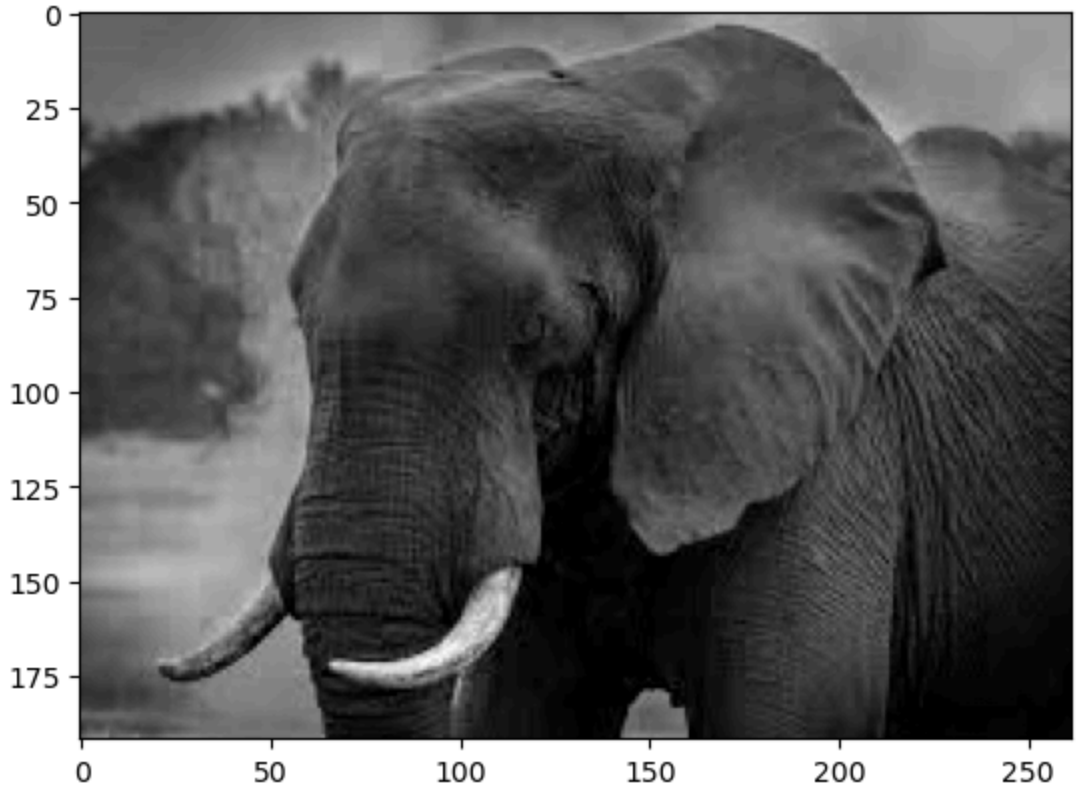
In [17]:
```
plt.imshow(ele_red[:,:,1],cmap='grey')
```

Out[17]:
```
<matplotlib.image.AxesImage at 0x2062e6e2710>
```



In [18]:
```
plt.imshow(ele_red[:,:,2],cmap='grey')
```

Out[18]:
```
<matplotlib.image.AxesImage at 0x2062f0fd450>
```



In [19]:
```
ele_red[:,:,0]
```

Out[19]:
```
array([[108, 108, 108, ..., 165, 164, 164],
       [108, 108, 108, ..., 165, 164, 164],
       [107, 107, 107, ..., 165, 164, 164],
       ...,
       [155, 169, 184, ...,   8,  10,  11],
       [190, 203, 217, ...,  10,  10,  10],
       [227, 236, 247, ...,  10,  10,   9]], shape=(192, 262), dtype=uint8)
```

In [20]:
```
ele_red[:,:,1]
```

Out[20]:
```
array([[107, 107, 107, ..., 160, 159, 159],
       [107, 107, 107, ..., 160, 159, 159],
       [106, 106, 106, ..., 160, 159, 159],
       ...,
       [103, 119, 134, ...,   8,  10,  11],
       [129, 142, 156, ...,  10,  10,  10],
       [161, 170, 181, ...,  10,  10,   9]], shape=(192, 262), dtype=uint8)
```

In [21]:
```
ele_red[:,:,2]
```

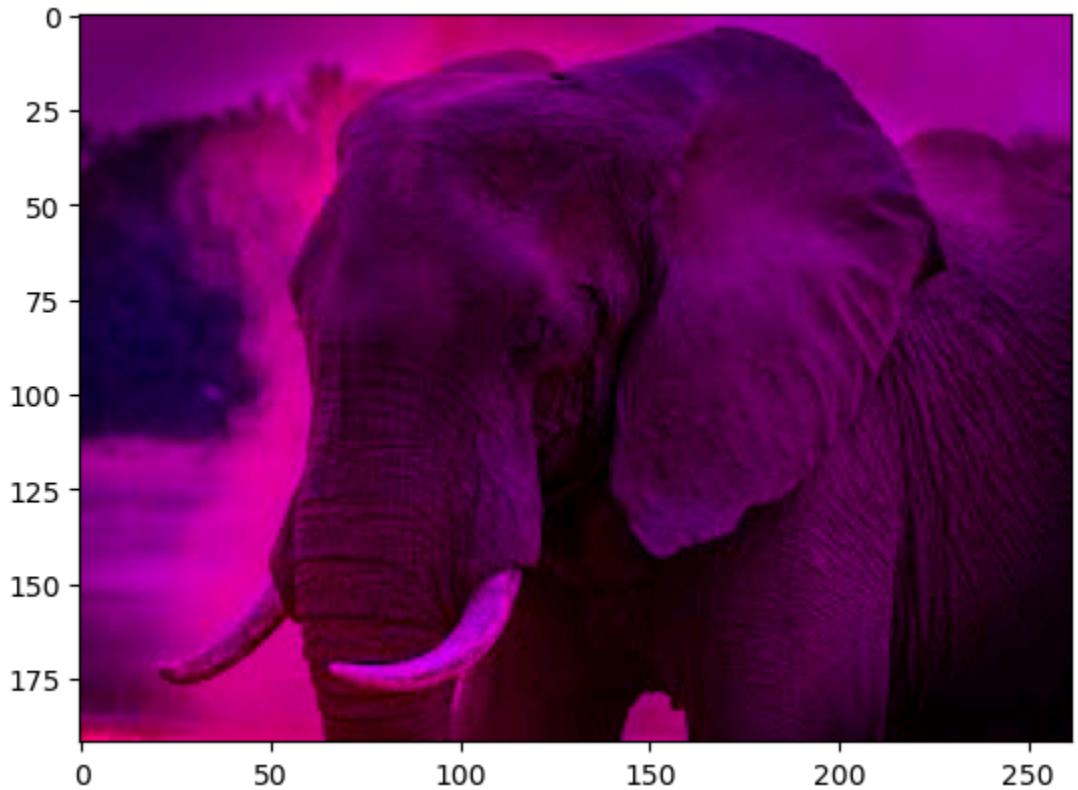
Out[21]:
```
array([[103, 103, 103, ..., 156, 155, 155],
       [103, 103, 103, ..., 156, 155, 155],
       [102, 102, 102, ..., 156, 155, 155],
       ...,
       [ 55,  70,  85, ...,   6,   8,   9],
       [ 72,  85,  99, ...,   8,   8,   8],
       [100, 109, 120, ...,   8,   8,   7]], shape=(192, 262), dtype=uint8)
```

In [23]:
```
ele_red[:,:,1]=0
```

In [ ]:
```
ele_red[:,:,1]
```

In [24]:
```
plt.imshow(ele_red)
```

Out[24]:
```
<matplotlib.image.AxesImage at 0x2062f1b3ed0>
```

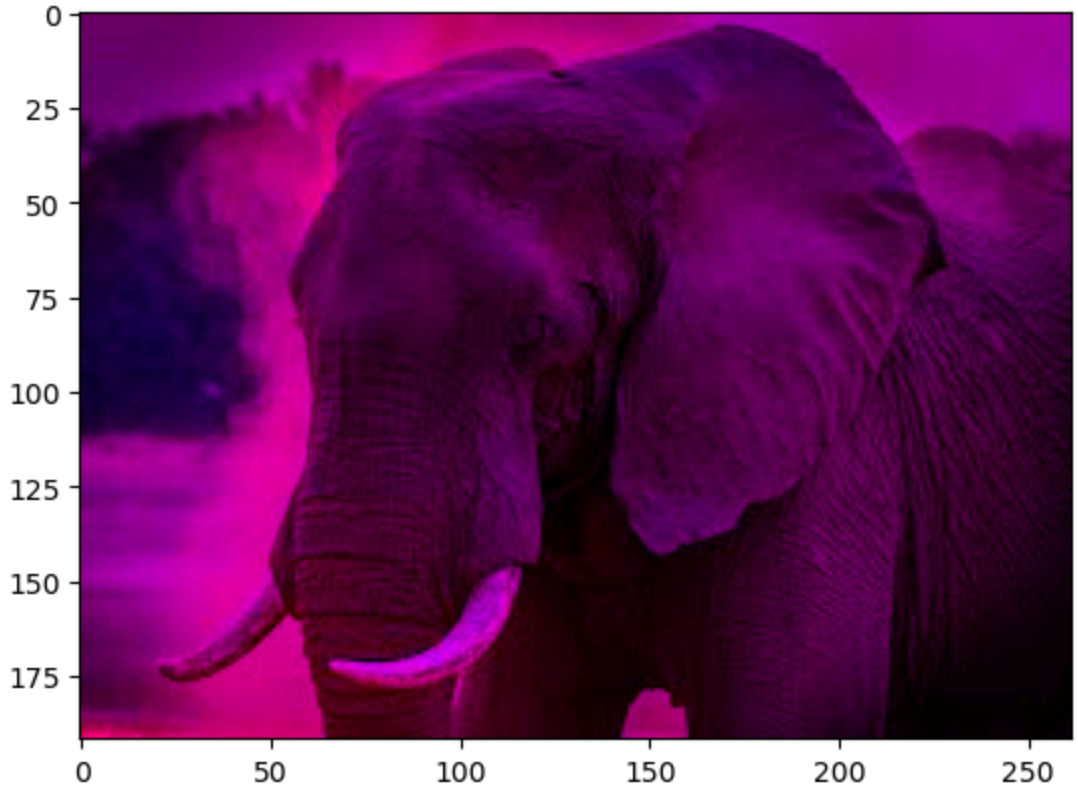


In [ ]:
```
ele_red[:,:,2]=0
```

In [ ]:
```
ele_red[:,:,2]  #The image becomes reddish.
```

In [25]:
```
plt.imshow(ele_red)
```

Out[25]:
```
<matplotlib.image.AxesImage at 0x2062f356c10>
```



In [26]:
```
ele_array
```

Out[26]:
```
array([[[108, 107, 103],
        [108, 107, 103],
        [108, 107, 103],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       [[108, 107, 103],
        [108, 107, 103],
        [108, 107, 103],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       [[107, 106, 102],
        [107, 106, 102],
        [107, 106, 102],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       ...,

       [[155, 103,  55],
        [169, 119,  70],
        [184, 134,  85],
        ...,
        [  8,   8,   6],
        [ 10,  10,   8],
        [ 11,  11,   9]],

       [[190, 129,  72],
        [203, 142,  85],
        [217, 156,  99],
        ...,
        [ 10,  10,   8],
        [ 10,  10,   8],
        [ 10,  10,   8]],

       [[227, 161, 100],
        [236, 170, 109],
        [247, 181, 120],
        ...,
        [ 10,  10,   8],
        [ 10,  10,   8],
        [  9,   9,   7]]], shape=(192, 262, 3), dtype=uint8)
```

In [27]:
```
ele_red
```

Out[27]:
```
array([[[108,   0, 103],
        [108,   0, 103],
        [108,   0, 103],
        ...,
        [165,   0, 156],
        [164,   0, 155],
        [164,   0, 155]],

       [[108,   0, 103],
        [108,   0, 103],
        [108,   0, 103],
        ...,
        [165,   0, 156],
        [164,   0, 155],
        [164,   0, 155]],

       [[107,   0, 102],
        [107,   0, 102],
        [107,   0, 102],
        ...,
        [165,   0, 156],
        [164,   0, 155],
        [164,   0, 155]],

       ...,

       [[155,   0,  55],
        [169,   0,  70],
        [184,   0,  85],
        ...,
        [  8,   0,   6],
        [ 10,   0,   8],
        [ 11,   0,   9]],

       [[190,   0,  72],
        [203,   0,  85],
        [217,   0,  99],
        ...,
        [ 10,   0,   8],
        [ 10,   0,   8],
        [ 10,   0,   8]],

       [[227,   0, 100],
        [236,   0, 109],
        [247,   0, 120],
        ...,
        [ 10,   0,   8],
        [ 10,   0,   8],
        [  9,   0,   7]]], shape=(192, 262, 3), dtype=uint8)
```

In [28]:
```
ele
```

Out[28]:

In [44]:
```
ele_red[:,:,0]
```

Out[44]:
```
array([[108, 108, 108, ..., 165, 164, 164],
       [108, 108, 108, ..., 165, 164, 164],
       [107, 107, 107, ..., 165, 164, 164],
       ...,
       [155, 169, 184, ...,   8,  10,  11],
       [190, 203, 217, ...,  10,  10,  10],
       [227, 236, 247, ...,  10,  10,   9]], shape=(192, 262), dtype=uint8)
```

In [45]:
```
ele_red[:,:,1]
```

Out[45]:
```
array([[107, 107, 107, ..., 160, 159, 159],
       [107, 107, 107, ..., 160, 159, 159],
       [106, 106, 106, ..., 160, 159, 159],
       ...,
       [103, 119, 134, ...,   8,  10,  11],
       [129, 142, 156, ...,  10,  10,  10],
       [161, 170, 181, ...,  10,  10,   9]], shape=(192, 262), dtype=uint8)
```

In [46]:
```
ele_red[:,:,2]
```

Out[46]:
```
array([[103, 103, 103, ..., 156, 155, 155],
       [103, 103, 103, ..., 156, 155, 155],
       [102, 102, 102, ..., 156, 155, 155],
       ...,
       [ 55,  70,  85, ...,   6,   8,   9],
       [ 72,  85,  99, ...,   8,   8,   8],
       [100, 109, 120, ...,   8,   8,   7]], shape=(192, 262), dtype=uint8)
```

In [54]:
```
ele_red[:,:,:]=0
ele_red
```

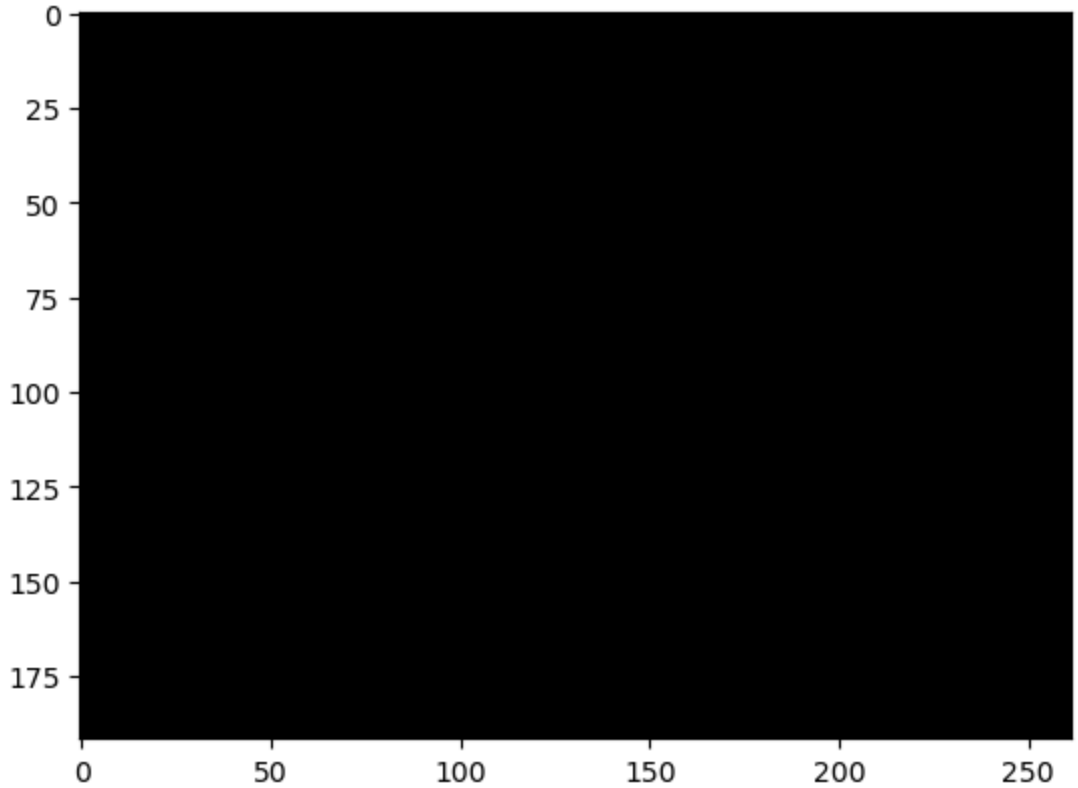
Out[54]:
```
array([[[0, 0, 0],
        [0, 0, 0],
        [0, 0, 0],
        ...,
        [0, 0, 0],
        [0, 0, 0],
        [0, 0, 0]],

       [[0, 0, 0],
        [0, 0, 0],
        [0, 0, 0],
        ...,
        [0, 0, 0],
        [0, 0, 0],
        [0, 0, 0]],

       [[0, 0, 0],
        [0, 0, 0],
        [0, 0, 0],
        ...,
        [0, 0, 0],
        [0, 0, 0],
        [0, 0, 0]],

       ...,

       [[0, 0, 0],
        [0, 0, 0],
        [0, 0, 0],
        ...,
        [0, 0, 0],
        [0, 0, 0],
        [0, 0, 0]],

       [[0, 0, 0],
        [0, 0, 0],
        [0, 0, 0],
        ...,
        [0, 0, 0],
        [0, 0, 0],
        [0, 0, 0]],

       [[0, 0, 0],
        [0, 0, 0],
        [0, 0, 0],
        ...,
        [0, 0, 0],
        [0, 0, 0],
        [0, 0, 0]]], shape=(192, 262, 3), dtype=uint8)
```

In [53]:
```
ele_red[:,:,0]=0
ele_red
```

Out[53]:
```
array([[[  0, 107, 103],
        [  0, 107, 103],
        [  0, 107, 103],
        ...,
        [  0, 160, 156],
        [  0, 159, 155],
        [  0, 159, 155]],

       [[  0, 107, 103],
        [  0, 107, 103],
        [  0, 107, 103],
        ...,
        [  0, 160, 156],
        [  0, 159, 155],
        [  0, 159, 155]],

       [[  0, 106, 102],
        [  0, 106, 102],
        [  0, 106, 102],
        ...,
        [  0, 160, 156],
        [  0, 159, 155],
        [  0, 159, 155]],

       ...,

       [[  0, 103,  55],
        [  0, 119,  70],
        [  0, 134,  85],
        ...,
        [  0,   8,   6],
        [  0,  10,   8],
        [  0,  11,   9]],

       [[  0, 129,  72],
        [  0, 142,  85],
        [  0, 156,  99],
        ...,
        [  0,  10,   8],
        [  0,  10,   8],
        [  0,  10,   8]],

       [[  0, 161, 100],
        [  0, 170, 109],
        [  0, 181, 120],
        ...,
        [  0,  10,   8],
        [  0,  10,   8],
        [  0,   9,   7]]], shape=(192, 262, 3), dtype=uint8)
```

In [55]:
```
plt.imshow(ele_red)
```

Out[55]:
```
<matplotlib.image.AxesImage at 0x206307d1950>
```



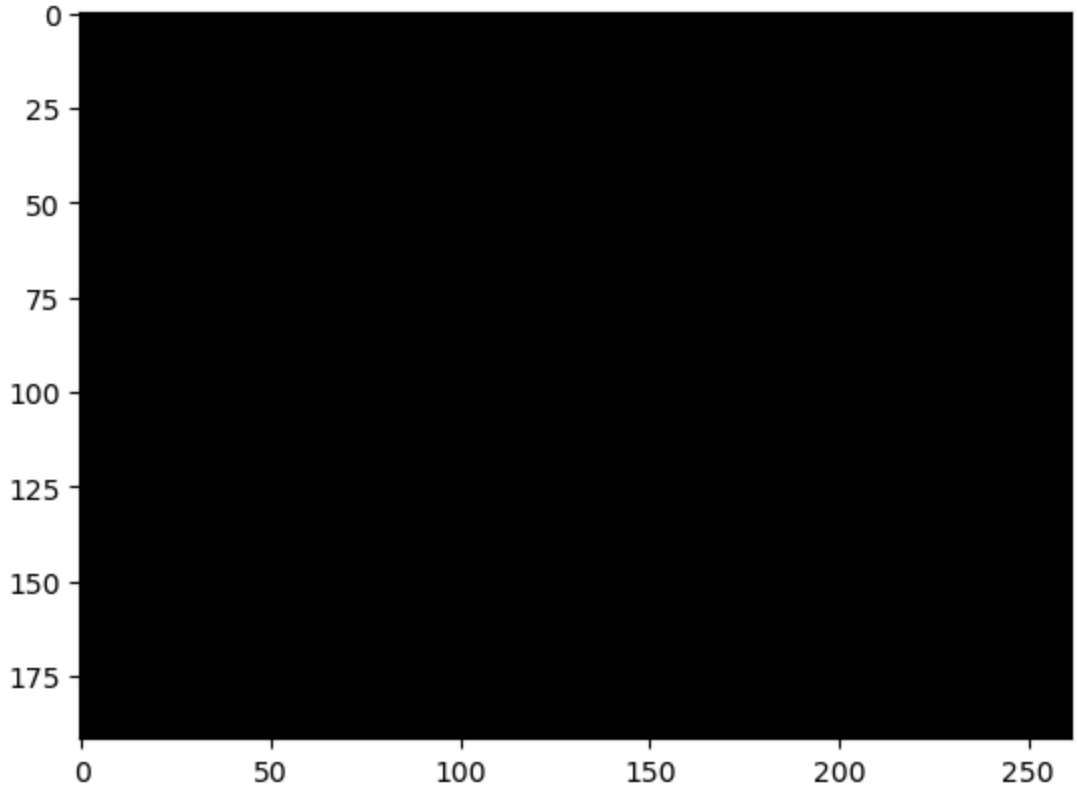
In [56]:
```
ele_red[:,:,1] = 0 #Set green to 0
```

In [57]:
```
ele_red[:,:,1] # H
```

Out[57]:
```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]], shape=(192, 262), dtype=uint8)
```

In [59]:
```
plt.imshow(ele_red)
#Now image become blank as in above code somewhere we have made red=0 and blue
```

Out[59]:
```
<matplotlib.image.AxesImage at 0x20630270050>
```



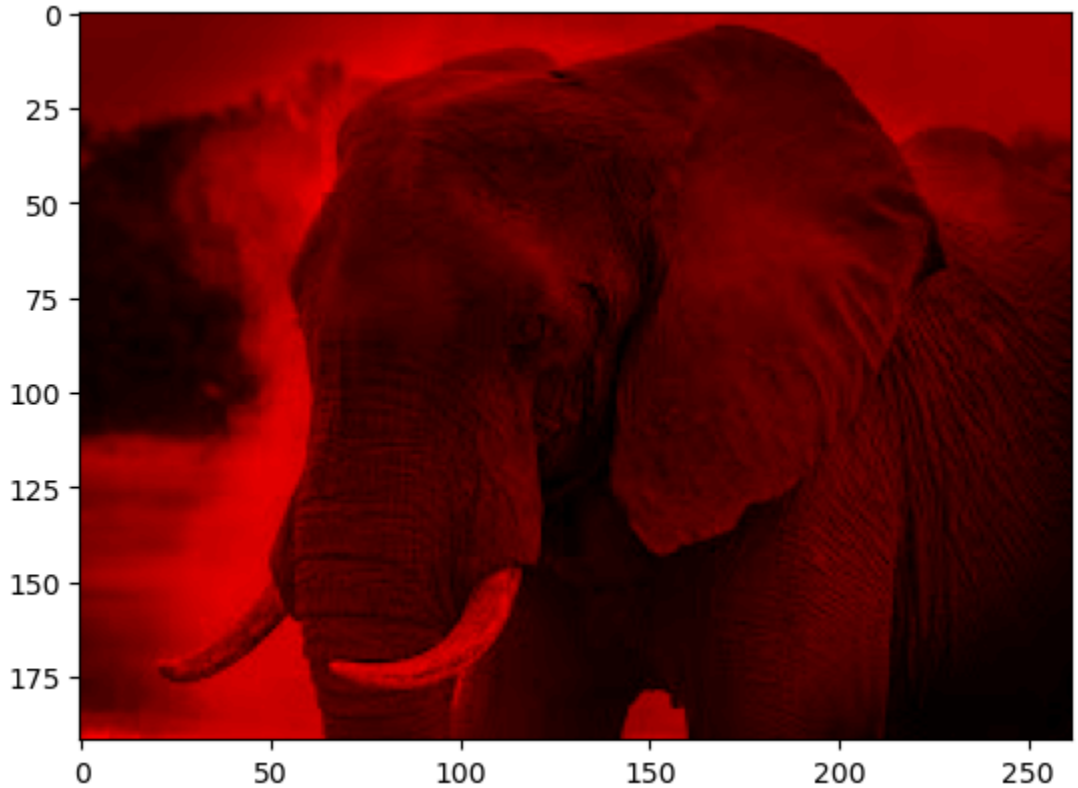
In [63]:
```
#To get back original value
ele_red = ele_array.copy()
ele_red
#ele_red[:,:,0] = 0   # remove red
#ele_red[:,:,1] = 0   # Remove green
#ele_red[:,:,2] = 0   # remove blue
```

Out[63]:
```
array([[[108, 107, 103],
        [108, 107, 103],
        [108, 107, 103],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       [[108, 107, 103],
        [108, 107, 103],
        [108, 107, 103],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       [[107, 106, 102],
        [107, 106, 102],
        [107, 106, 102],
        ...,
        [165, 160, 156],
        [164, 159, 155],
        [164, 159, 155]],

       ...,

       [[155, 103,  55],
        [169, 119,  70],
        [184, 134,  85],
        ...,
        [  8,   8,   6],
        [ 10,  10,   8],
        [ 11,  11,   9]],

       [[190, 129,  72],
        [203, 142,  85],
        [217, 156,  99],
        ...,
        [ 10,  10,   8],
        [ 10,  10,   8],
        [ 10,  10,   8]],

       [[227, 161, 100],
        [236, 170, 109],
        [247, 181, 120],
        ...,
        [ 10,  10,   8],
        [ 10,  10,   8],
        [  9,   9,   7]]], shape=(192, 262, 3), dtype=uint8)
```

In [66]:
```
# Red Image
red_img = ele_array.copy()
red_img[:,:,1] = 0
```

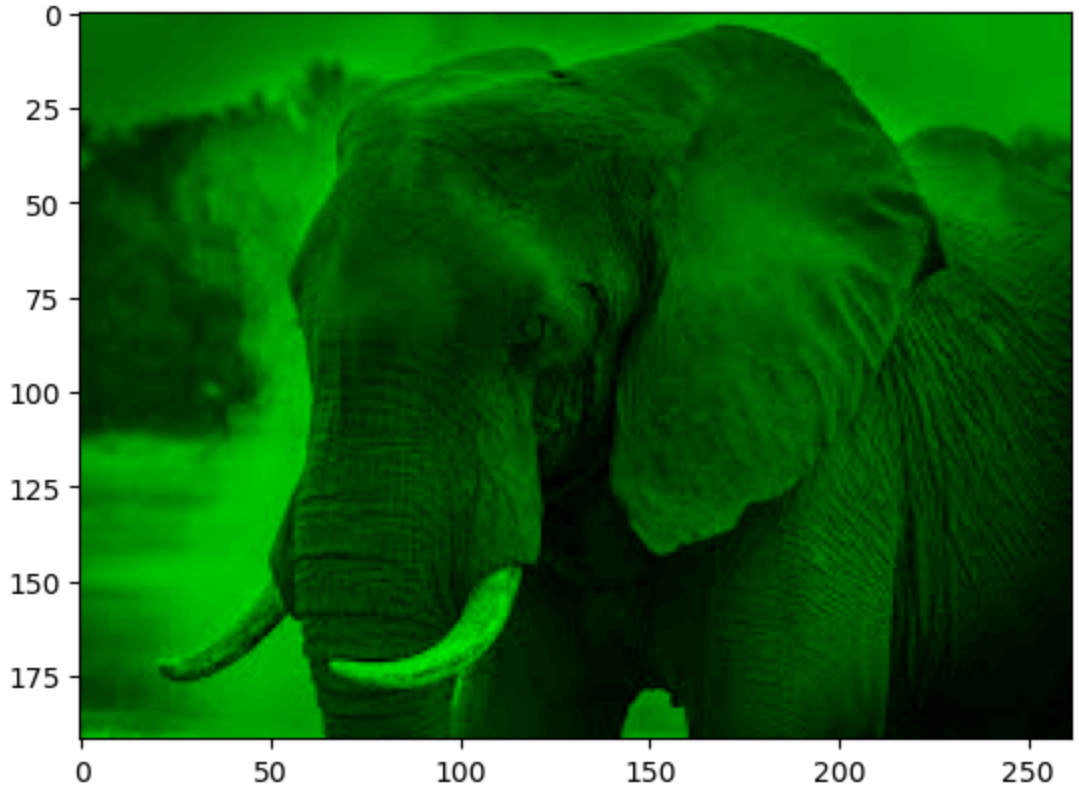
```
red_img[:,:,2] = 0
plt.imshow(red_img)
```

Out[66]: <matplotlib.image.AxesImage at 0x206303d6ad0>



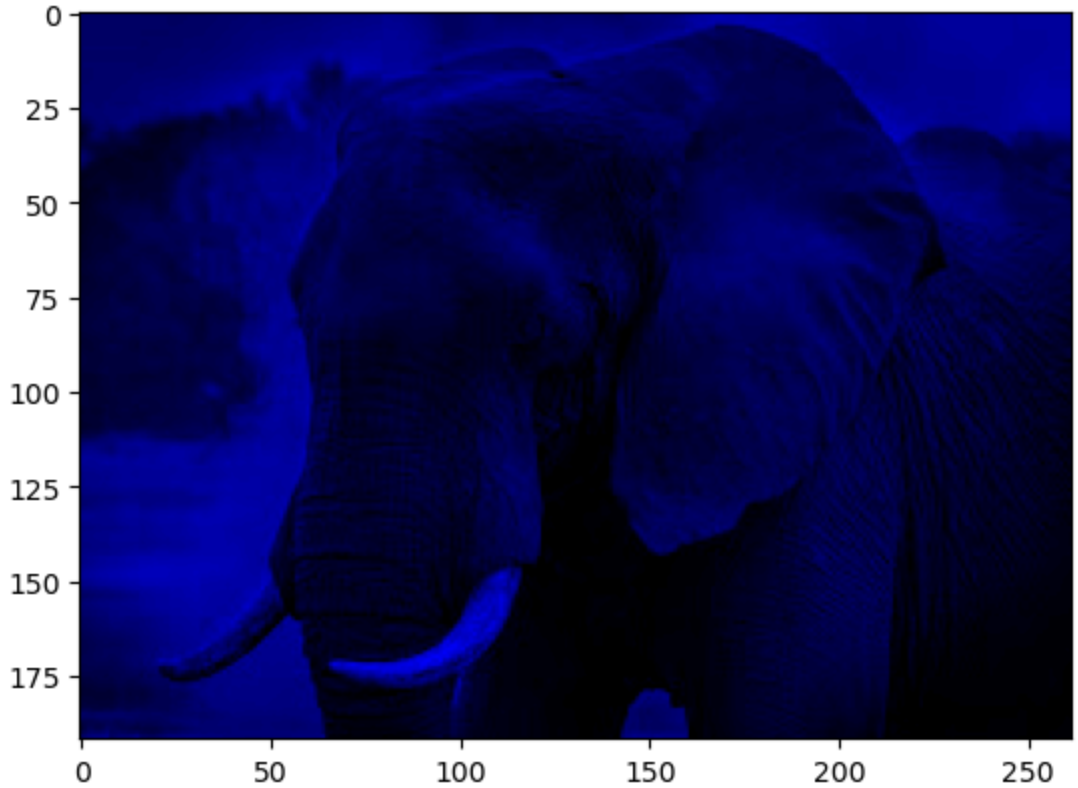
In [67]:
```
# Green Image
red_img = ele_array.copy()
red_img[:,:,0] = 0
red_img[:,:,2] = 0
plt.imshow(red_img)
```

Out[67]: <matplotlib.image.AxesImage at 0x206303e5810>



In [68]:
```
# Blue Image
red_img = ele_array.copy()
red_img[:,:,0] = 0
red_img[:,:,1] = 0
plt.imshow(red_img)
```

Out[68]:
```
<matplotlib.image.AxesImage at 0x2063044c550>
```



In [ ]:

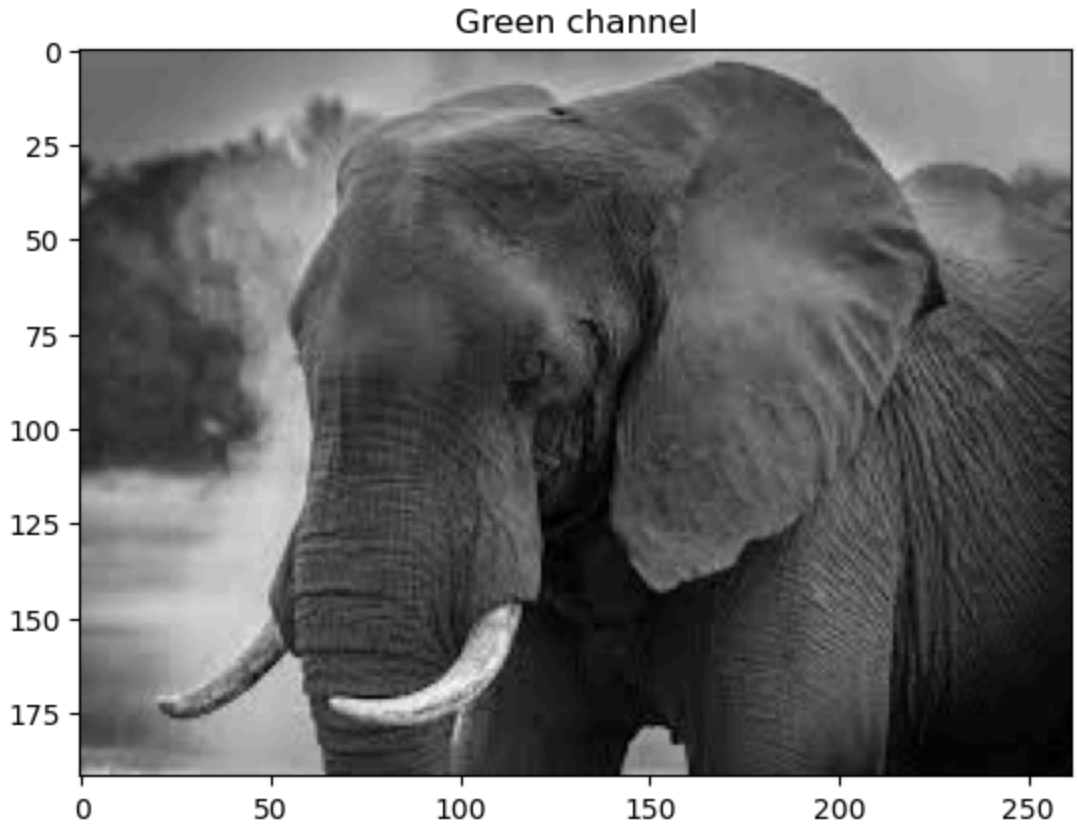
In [ ]:

==========

Code Result [:,:,0] Red channel [:,:,1] Green channel [:,:,2] Blue channel [:,:,1] = 0 Remove green [:,:,2] = 0 Remove blue both = 0 Only red remains

In [42]:
```
plt.imshow(ele_red[:,:,0], cmap='gray')  # Red
plt.title("Red channel")

plt.imshow(ele_red[:,:,2], cmap='gray')  # Blue
plt.title("Blue channel")

# to restore green channel
ele_red[:,:,1] = ele_array[:,:,1]

plt.imshow(ele_red[:,:,1], cmap='gray')  # Green
plt.title("Green channel")
```

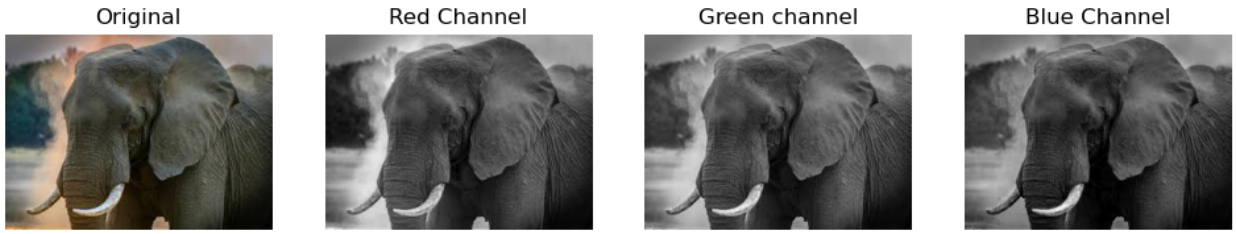
Out[42]:
```
Text(0.5, 1.0, 'Green channel')
```



In [ ]:

In [43]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(12,4))

# Original image
plt.subplot(1,4,1)
plt.imshow(ele_red)
plt.title("Original")
plt.axis('off')

# Red channel
plt.subplot(1,4,2)
plt.imshow(ele_red[:,:,0], cmap='gray')
plt.title("Red Channel")
plt.axis('off')

# Green channel
plt.subplot(1,4,3)
plt.imshow(ele_red[:,:,1], cmap='gray')
#plt.imshow(ele_array[:,:,1], cmap='gray') # used this code to restore green a
plt.title("Green channel")
plt.axis('off')

# Blue channel
plt.subplot(1,4,4)
```

```
plt.imshow(ele_red[:,:,2], cmap='gray')
plt.title("Blue Channel")
plt.axis('off')

plt.show()
```



In [ ]: